

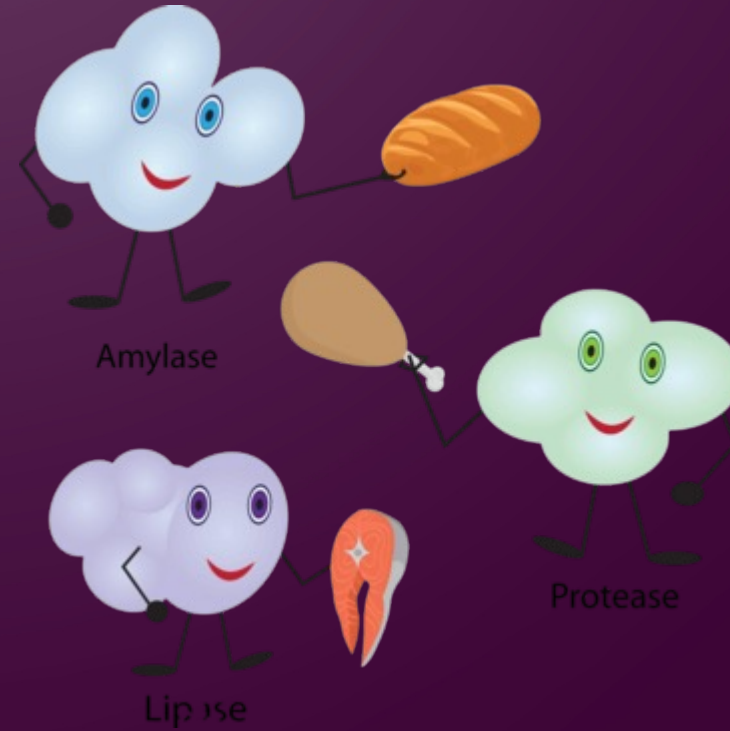


The M-CATs

SEMINAR 5

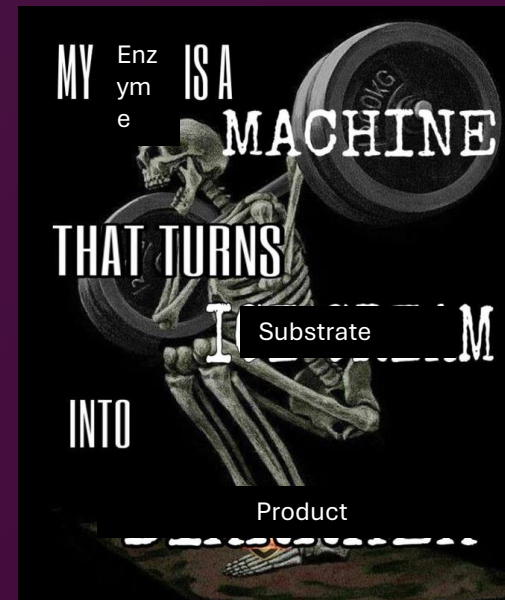
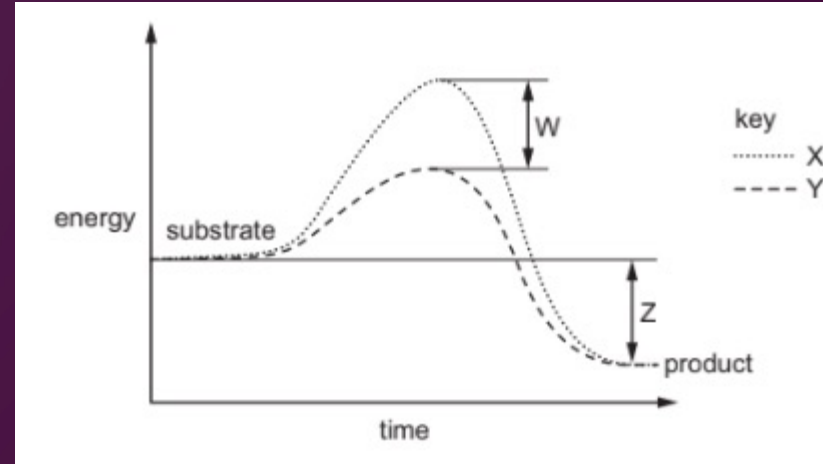
Biochemistry Chapter II

Enzymes



What Is an Enzyme?

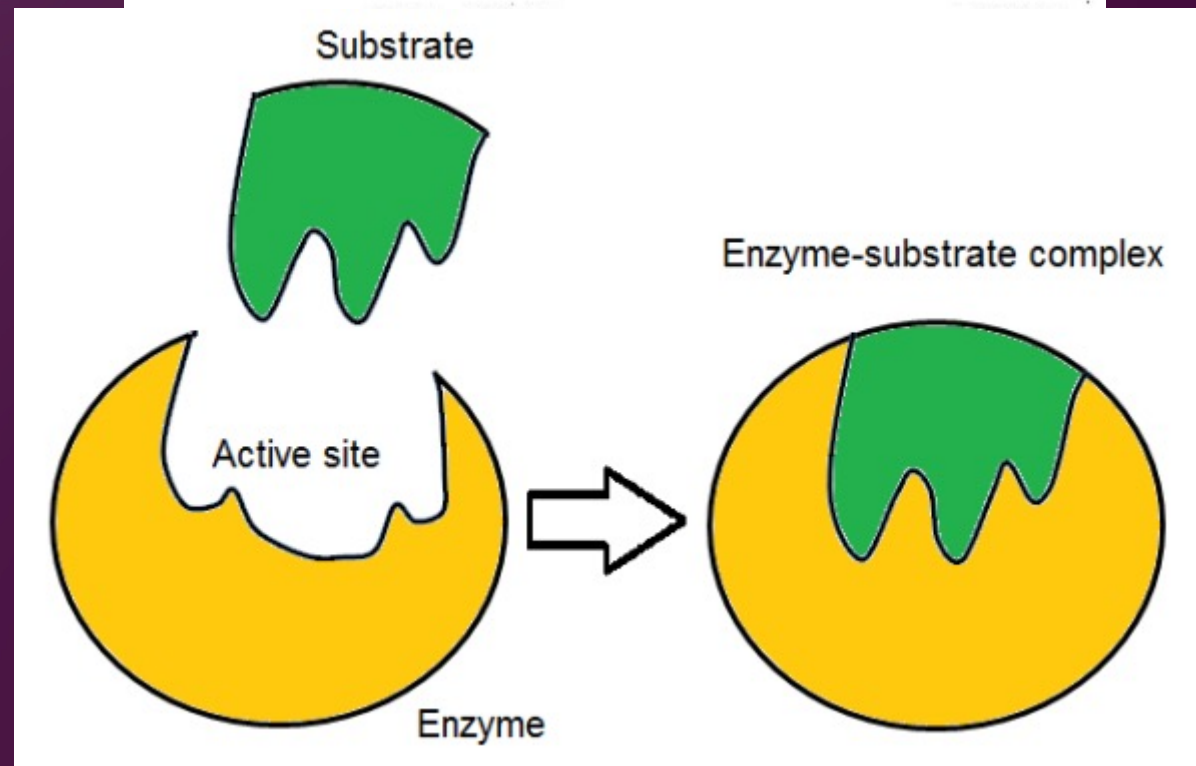
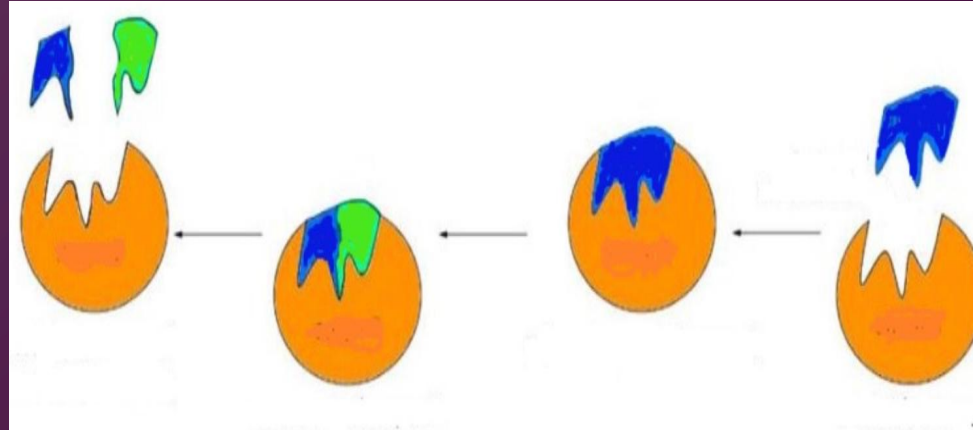
- Biological catalyst
- Not consumed
- Lower E_a
- NO effect on eq, ΔG , ΔH
- Activity depends on pH, T
- Specific function
- Speed up rxn bigtime



2.1: Enzymes as Biological Catalysts



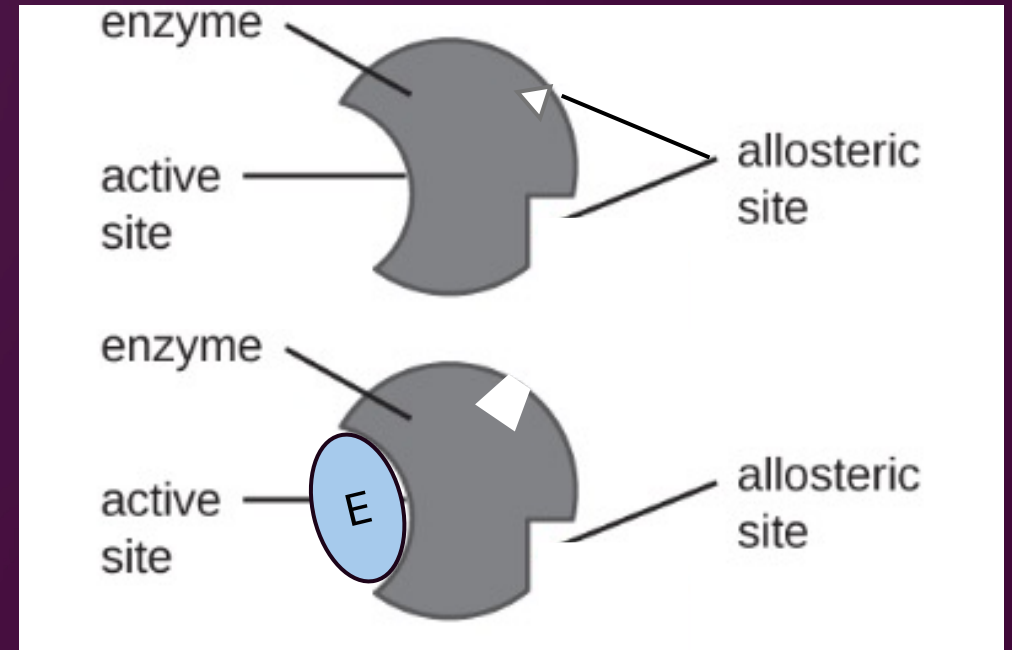
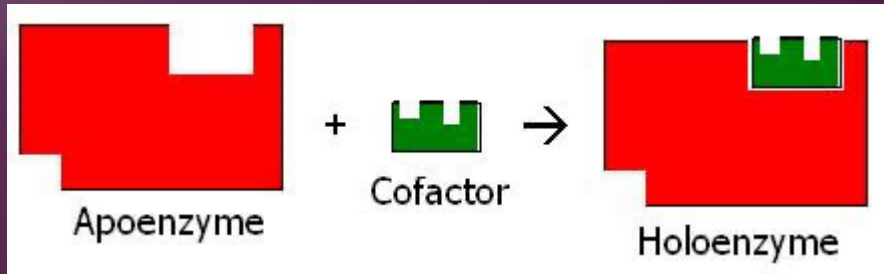
Anatomy of an Enzyme



2.2: Mechanisms of Enzyme Activity



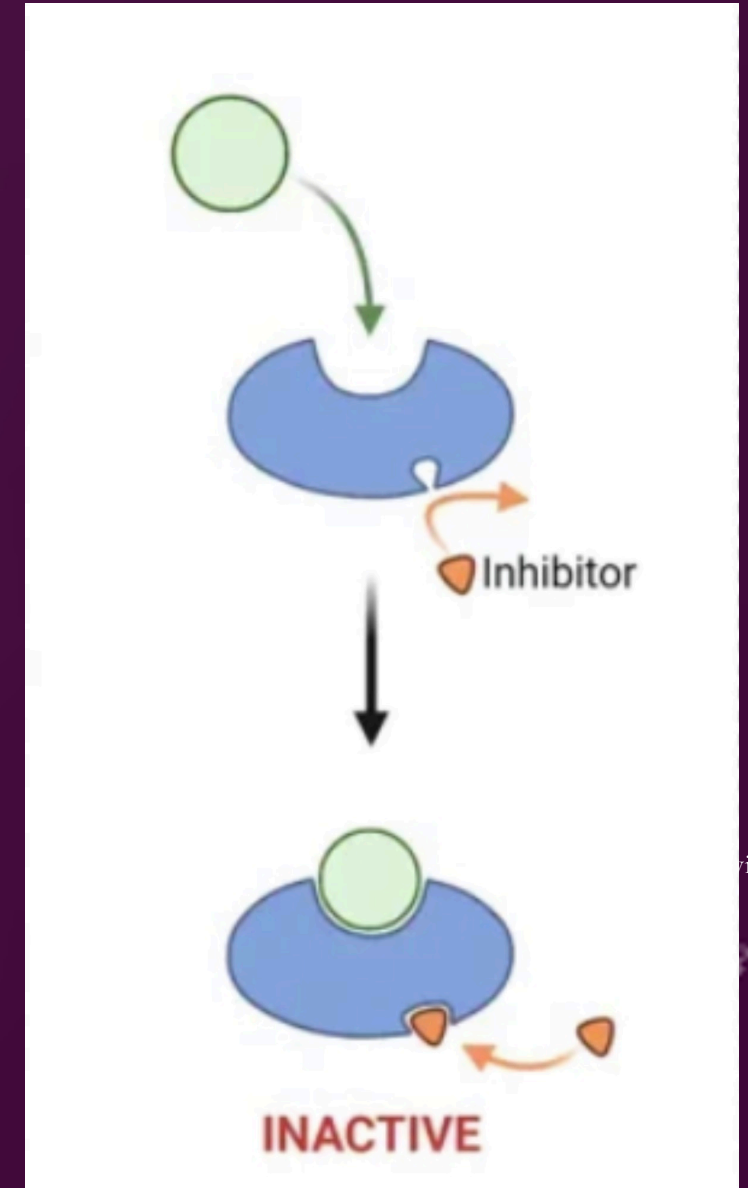
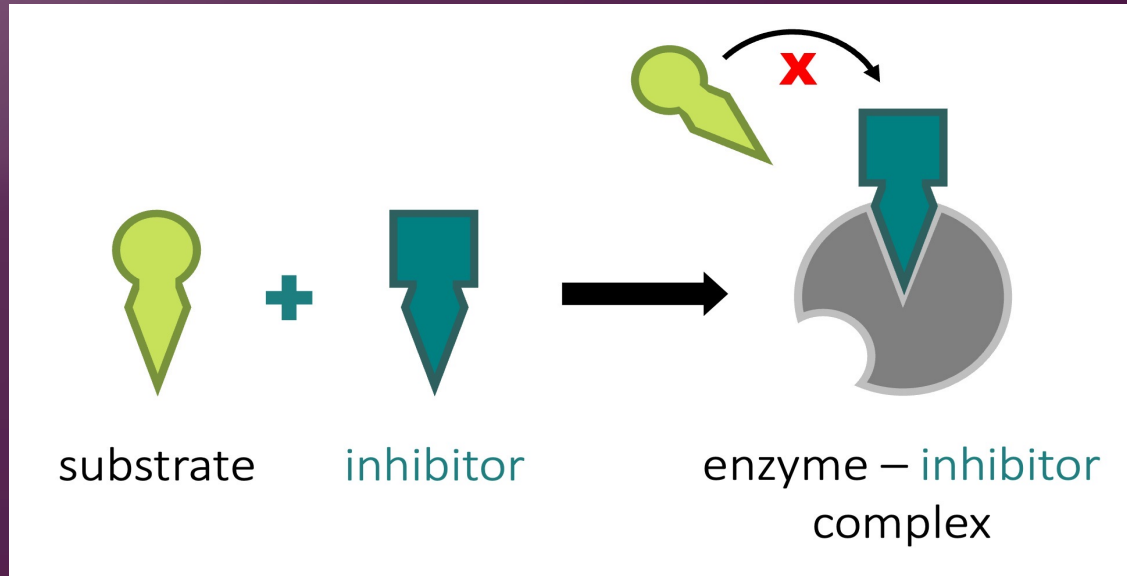
Anatomy of an Enzyme (cont.)



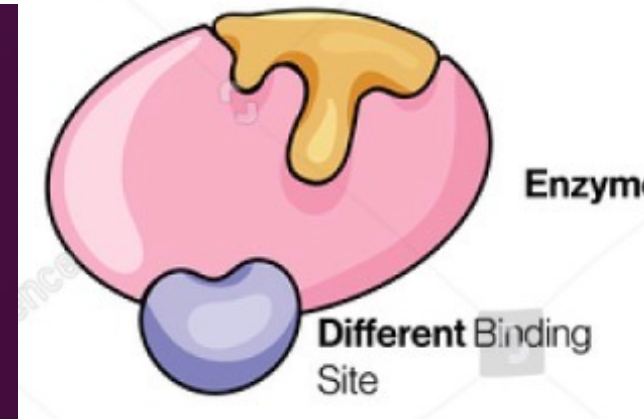
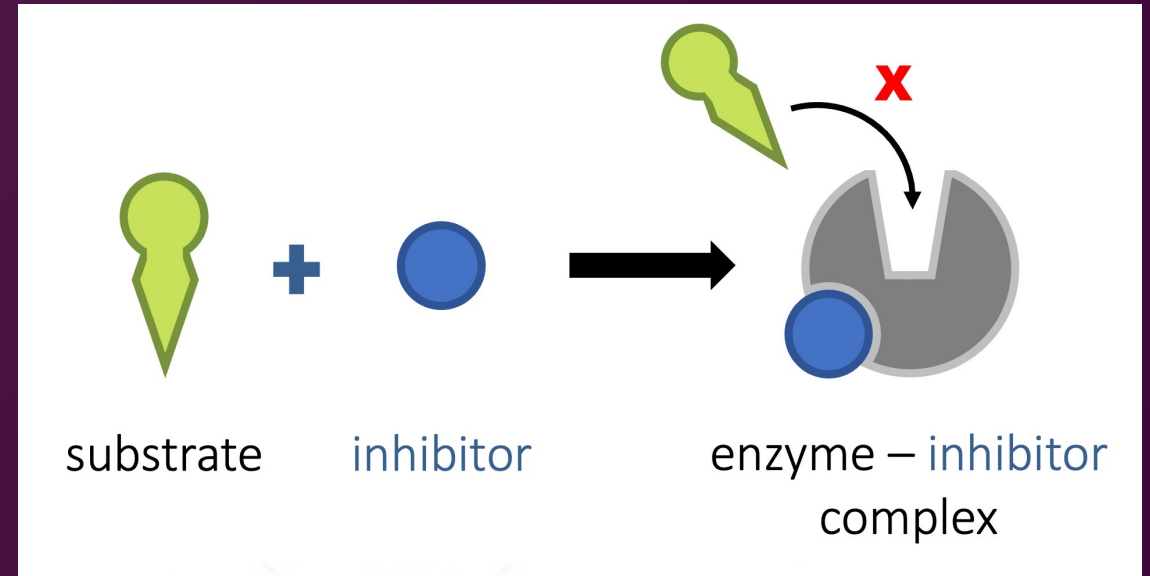
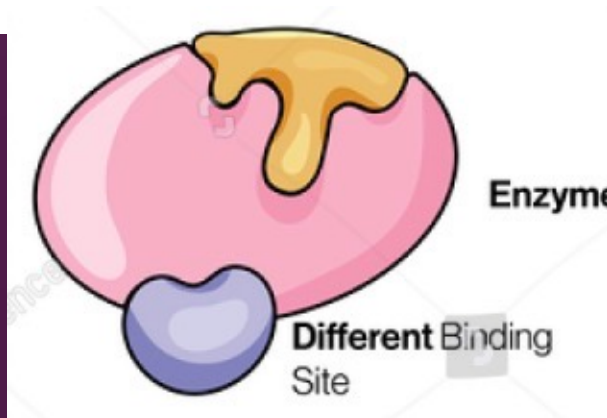
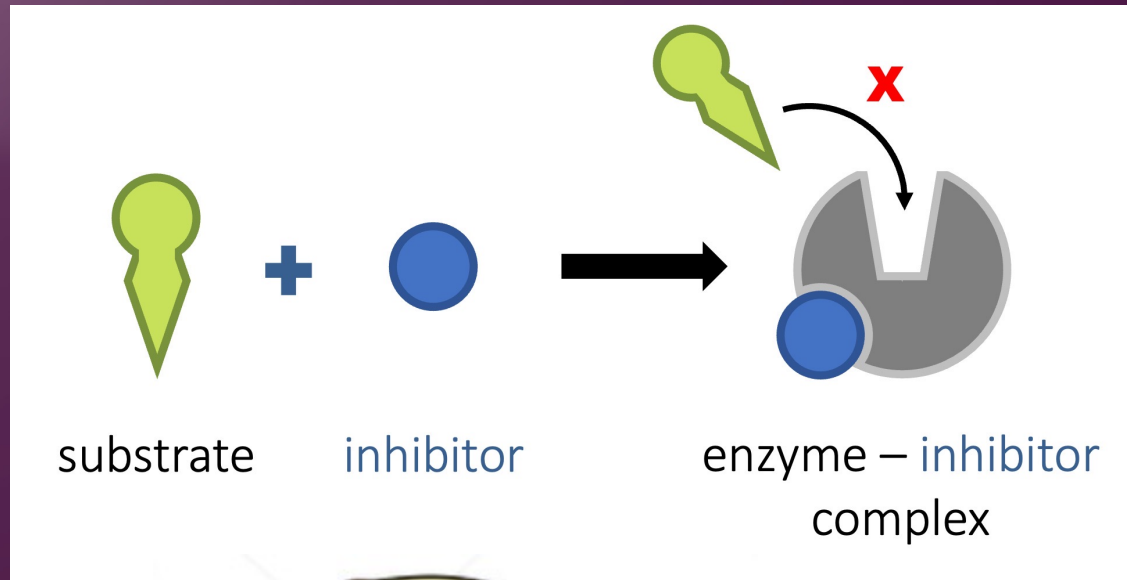
2.2: Mechanisms of Enzyme Activity



Reversible Inhibitors



Reversible Inhibitors

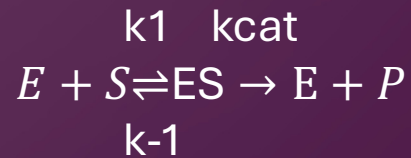


2.2: Mechanisms of Enzyme Activity



Enzyme Kinetics

(Michaelis-Menten Kinetics)



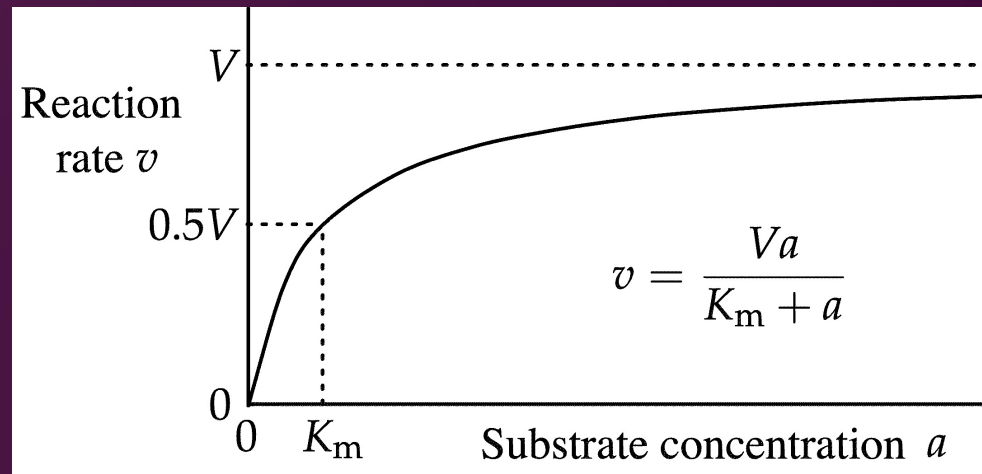
$$v = \frac{V_{max} [S]}{K_m + [S]}$$

$$V_{max} = [E]k_{cat}, v = \frac{[E]k_{cat}[S]}{K_m + [S]}$$

$$K_m \gg [S]: v = \frac{k_{cat}}{K_m} [E][S]$$

M-M experimental conditions:

1. v_0 ($[P]=0$)
2. Steady state approximation
3. Free ligand approximation
4. E only exists as E or ES



2.3: Enzyme Kinetics



Question

An enzyme catalyzes the conversion of substrate S to product P . Under a particular set of conditions, the enzyme has a V_{max} of 120 $\mu\text{M}/\text{min}$ and a K_m of 4.0 mM.

A researcher measures the initial reaction velocity at several substrate concentrations:

[S] (mM)	Initial velocity ($\mu\text{M}/\text{min}$)
0.5	13.3
1.0	24.0
2.0	40.0
4.0	60.0
8.0	80.0

Which statement best explains why the reaction velocity does not increase proportionally with substrate concentration?

- A.** The enzyme becomes denatured as substrate concentration increases.
- B.** Increasing substrate concentration causes the enzyme's K_m to increase.
- C.** The enzyme becomes increasingly saturated with substrate.
- D.** The equilibrium constant for the reaction decreases at high substrate concentration



Question

An enzyme-catalyzed reaction is performed with a substrate concentration that is much greater than the enzyme's K_m .

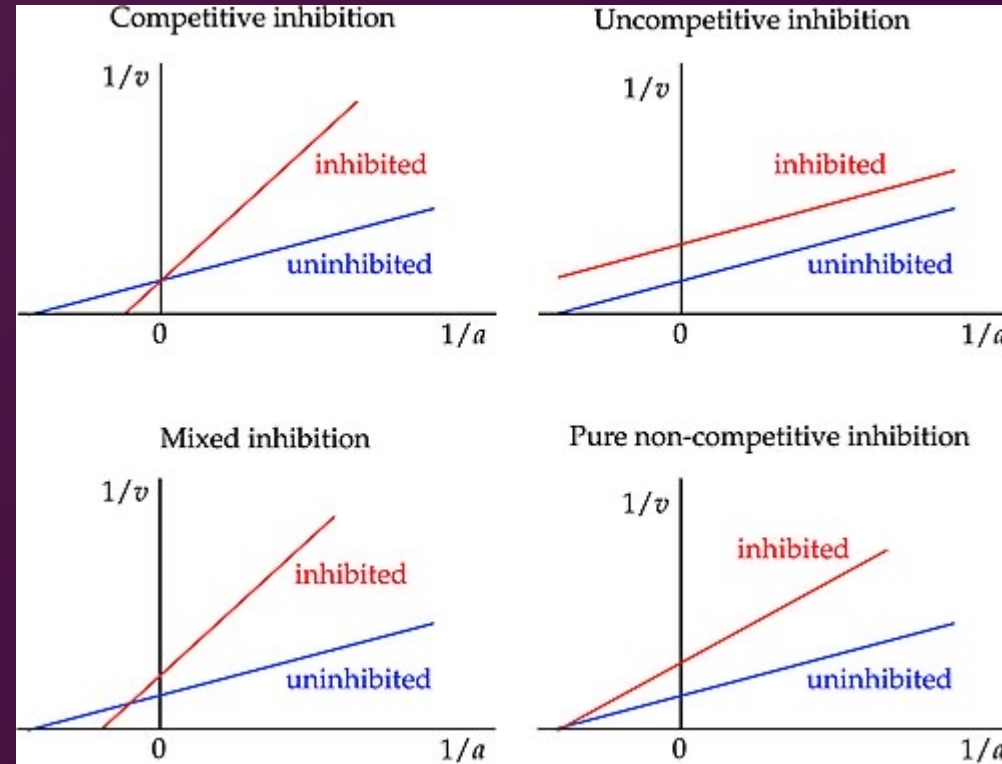
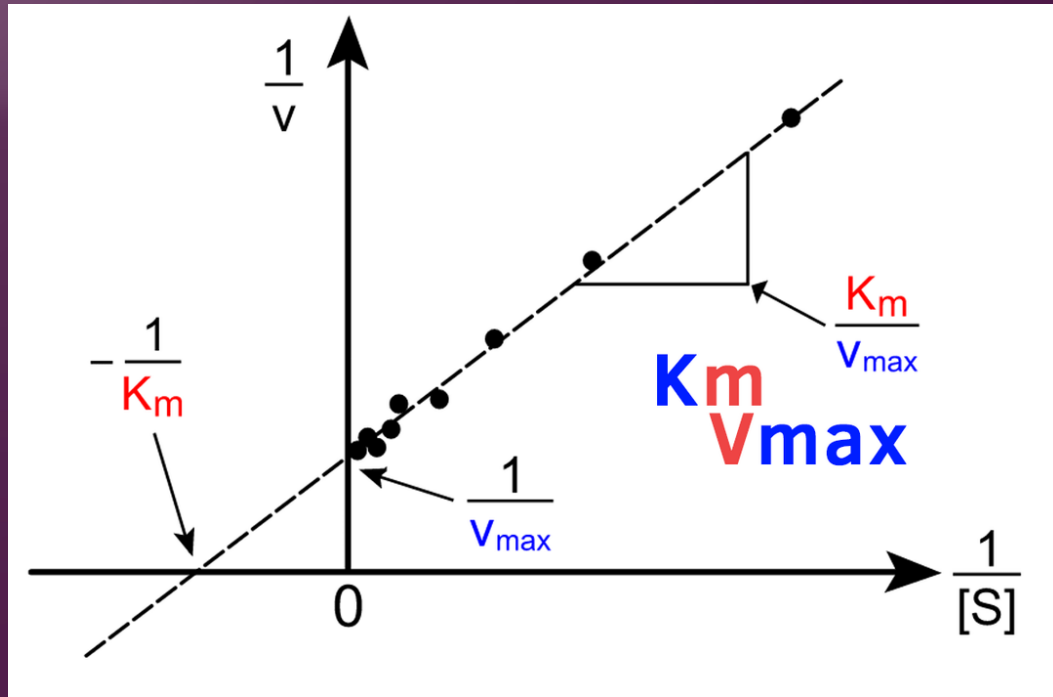
The researcher doubles the concentration of enzyme while keeping all other conditions constant. What is the expected effect?

- A. K_m doubles and V_{max} remains constant.
- B. K_m remains constant and V_{max} doubles.
- C. Both K_m and V_{max} double.
- D. Neither K_m nor V_{max} changes.



Enzyme Kinetics

(Lineweaver-Burk)

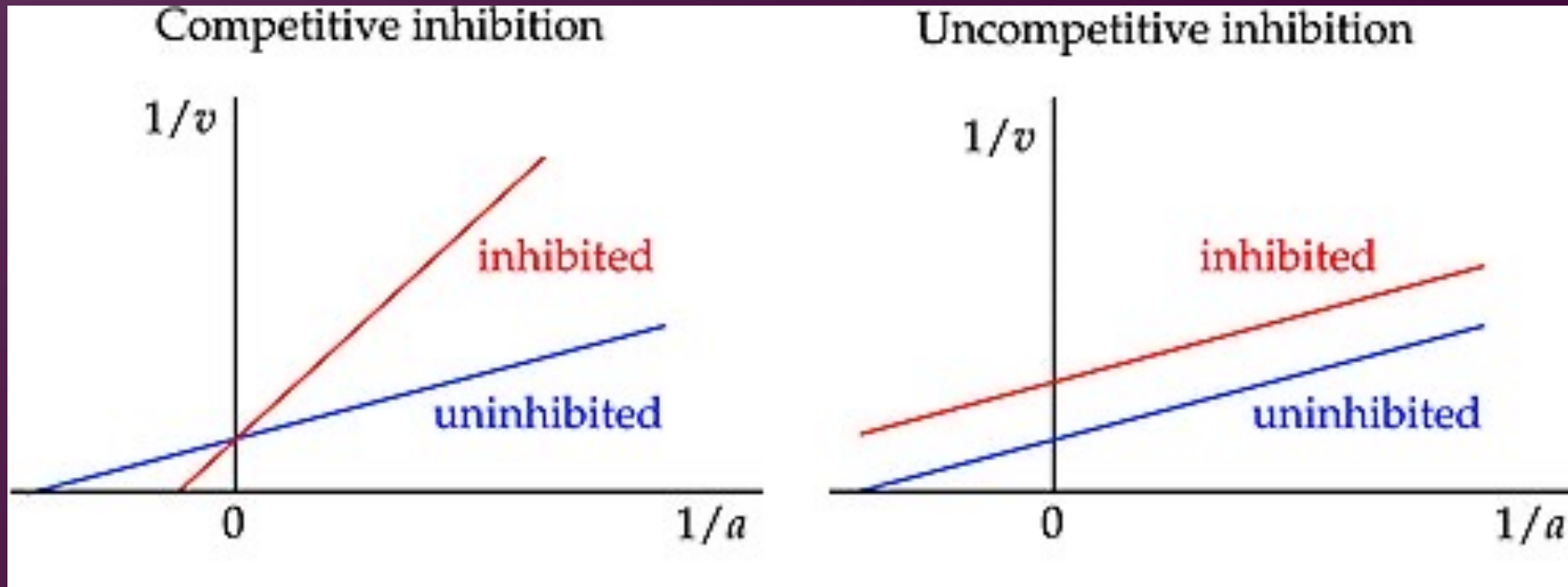


2.3, 2.5: Enzyme Kinetics, regulation



Enzyme Kinetics

(Lineweaver-Burk)

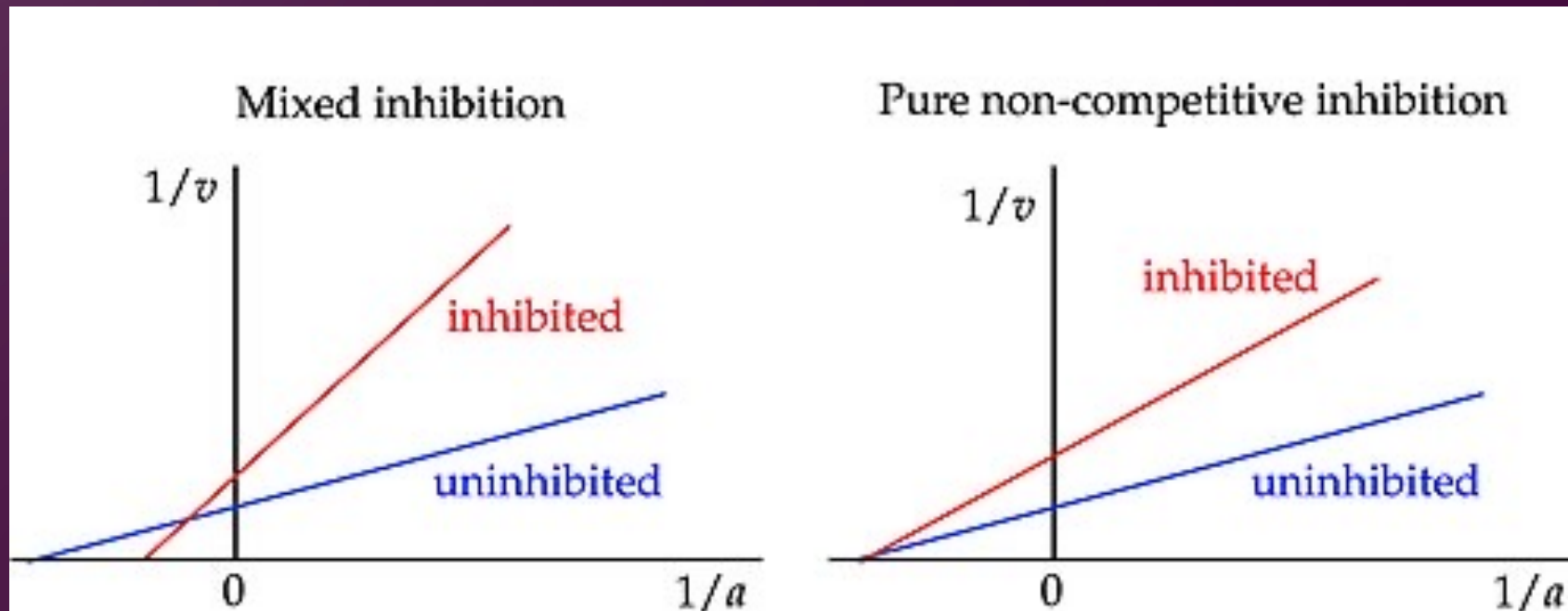


2.3, 2.5: Enzyme Kinetics, regulation



Enzyme Kinetics

(Lineweaver-Burk)



2.3, 2.5: Enzyme Kinetics,
regulation



Question

An investigator is studying the kinetic profile of an enzyme-catalyzed reaction in the presence of an unidentified reversible inhibitor, **X**. When plotting the data on a **Lineweaver-Burk (double-reciprocal) plot**, the investigator observes that as the concentration of inhibitor X increases, the resulting lines remain **strictly parallel** to the uninhibited enzyme line, shifting both intercepts further to the left and upward.

Which of the following best describes the binding preference and kinetic consequences of inhibitor X?

- A)** Inhibitor X binds exclusively to the free enzyme, decreasing V_{max} while leaving K_m unchanged.
- B)** Inhibitor X binds with equal affinity to both the free enzyme and the ES complex, reducing V_{max} without altering K_m .
- C)** Inhibitor X binds to the active site and can be outcompeted by high substrate concentrations, increasing apparent K_m while keeping V_{max} constant.
- D)** Inhibitor X binds exclusively to the enzyme-substrate (ES) complex, decreasing the inverse of both V_{max} and K_m proportionally.



Question

An enzyme follows Michaelis–Menten kinetics. In the absence of an inhibitor:

$$K_m = 2.0 \text{ mM}$$
$$V_{max} = 100 \text{ } \mu\text{M/min}$$

After addition of compound X, the researcher obtains:

$$K_m = 8.0 \text{ mM}$$
$$V_{max} = 100 \text{ } \mu\text{M/min}$$

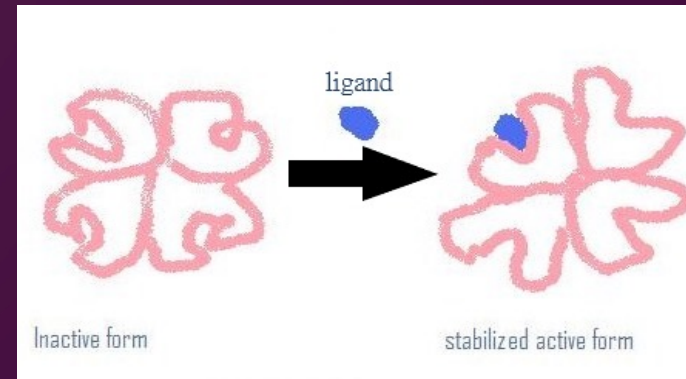
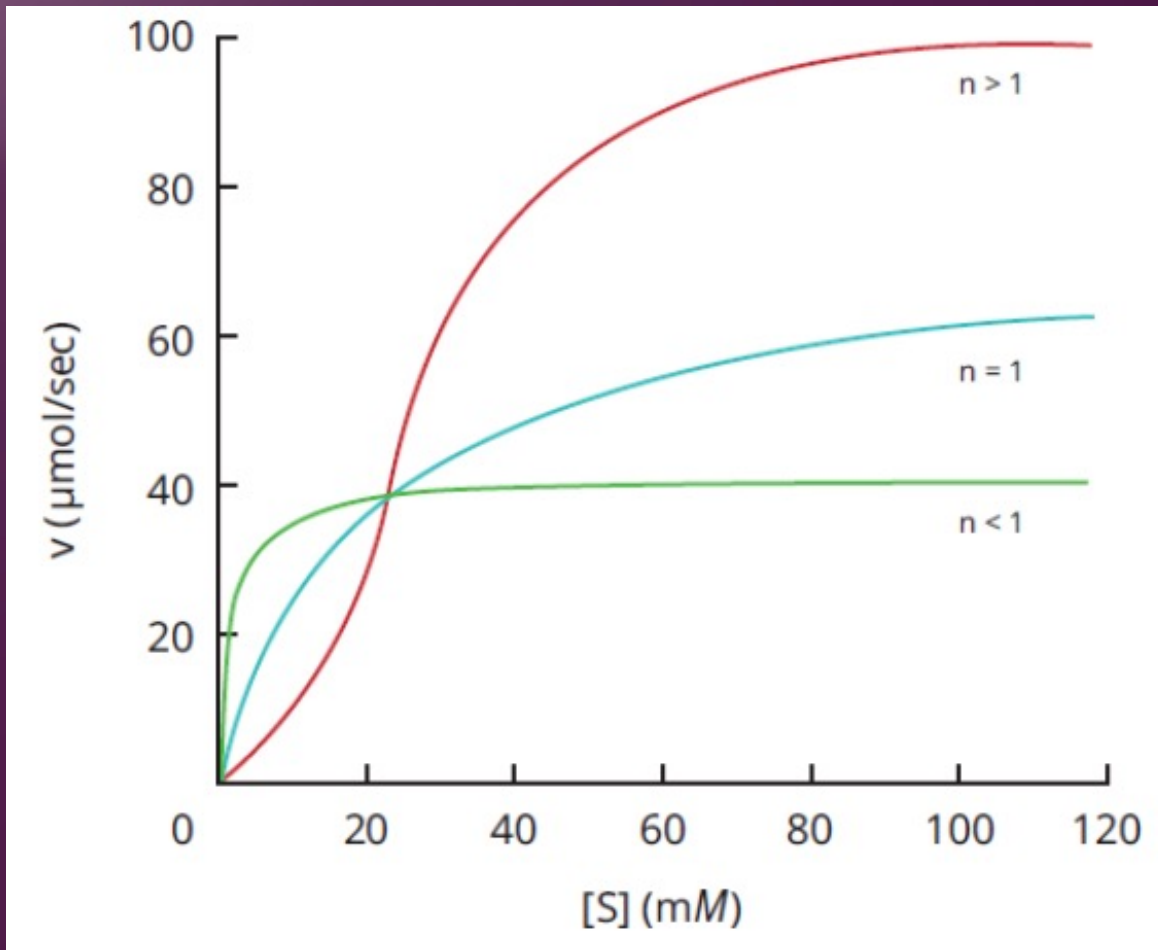
Which mechanism most likely explains the effect of compound X?

- A.** X binds irreversibly to the enzyme and prevents catalysis.
- B.** X binds to the active site and competes with substrate.
- C.** X binds to an allosteric site and decreases the amount of active enzyme.
- D.** X binds only to the enzyme–substrate complex.



Enzyme Kinetics

(Cooperativity)

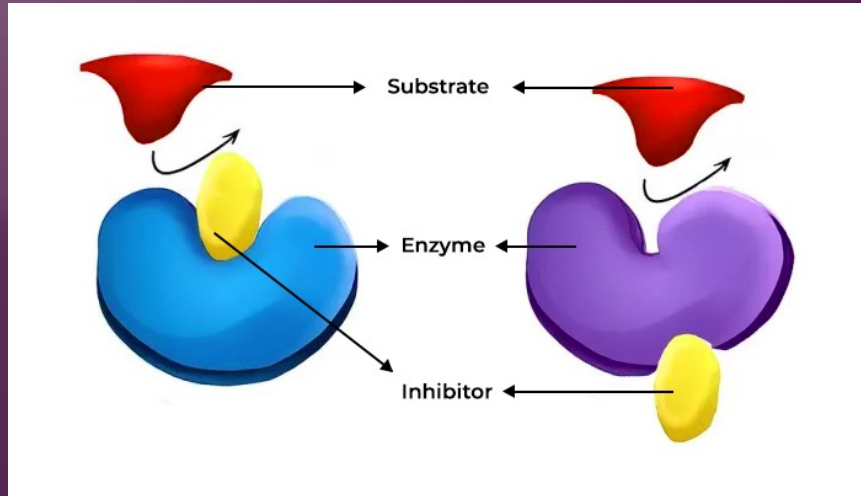


2.3: Enzyme Kinetics

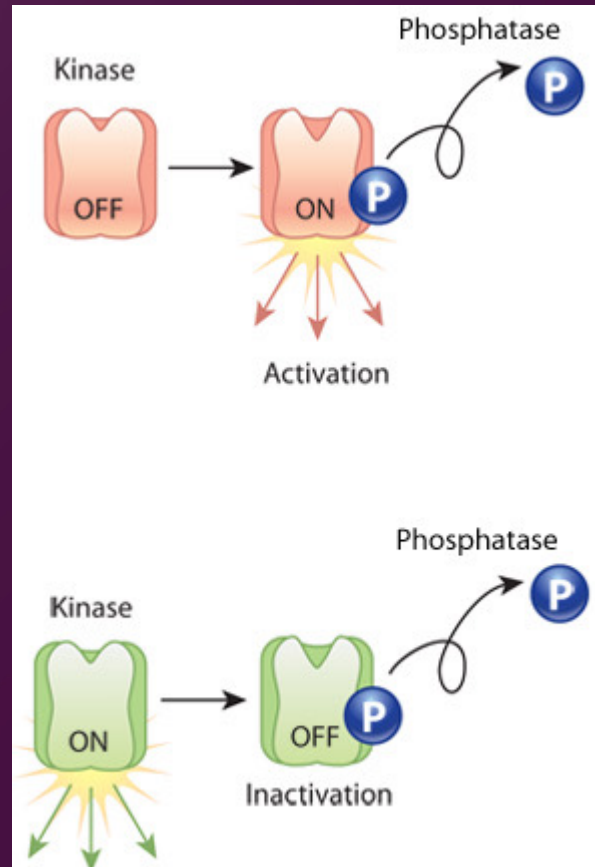


Irreversible Inhibition

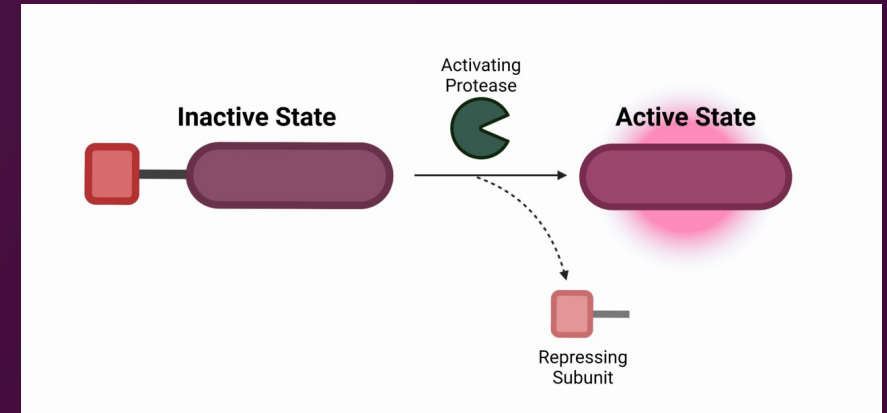
Allosteric regulators



Covalent Mods



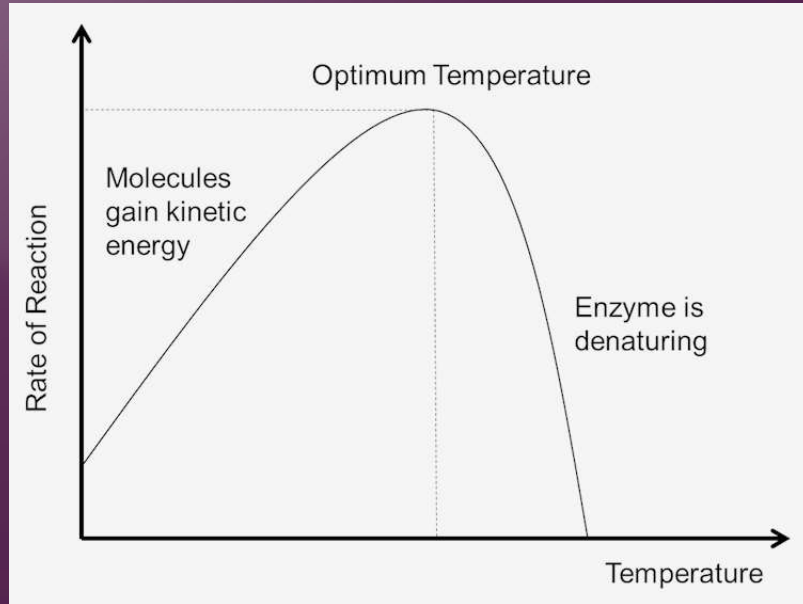
Zymogen



2.5: regulators



Other Factors in Enzyme Activity

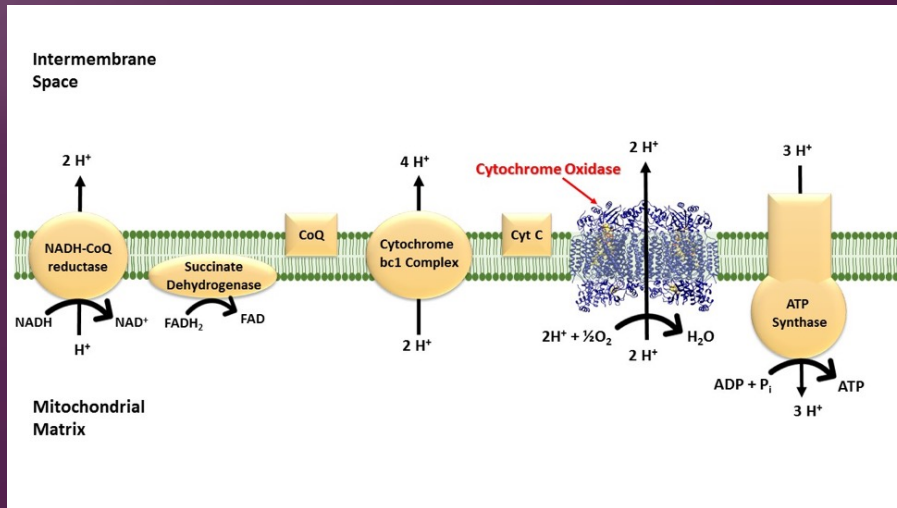


2.4: local conditions

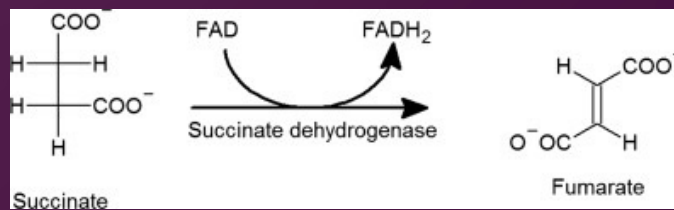
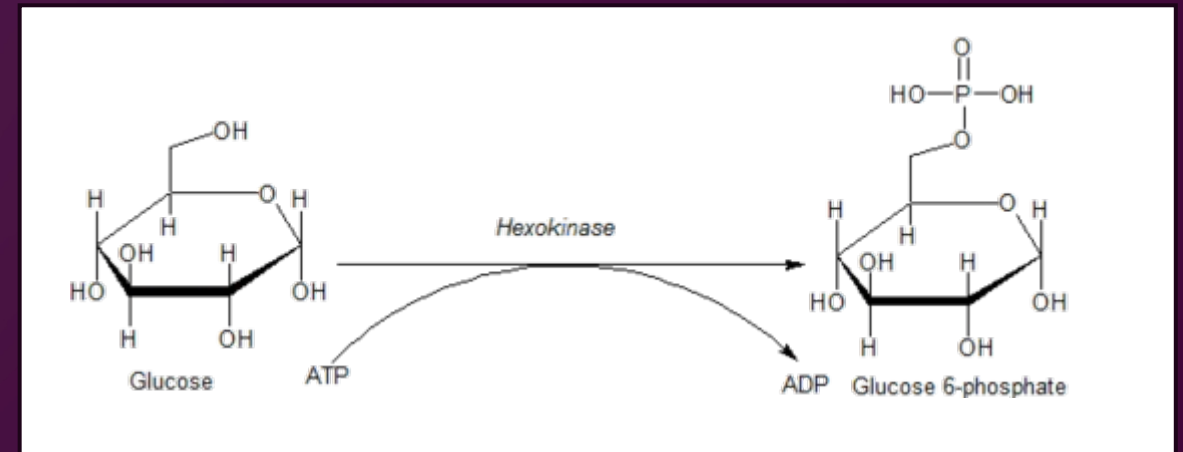


7 types

Oxidoreductases



Transferases

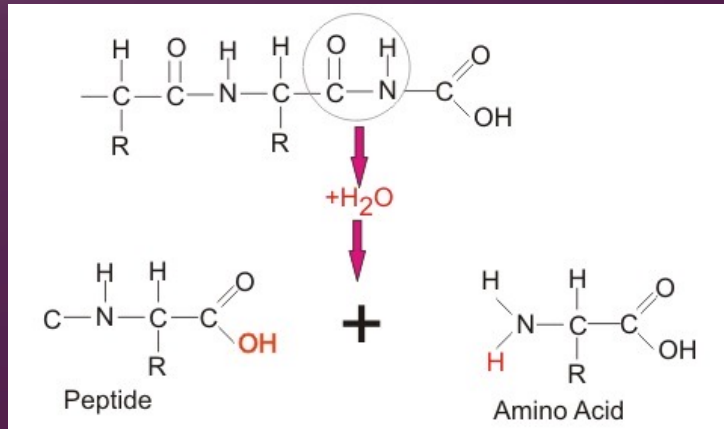


2.1: Enzymes as Biological Catalysts

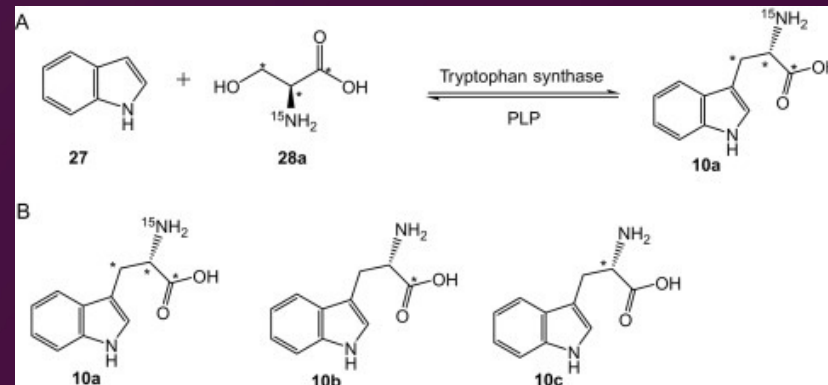
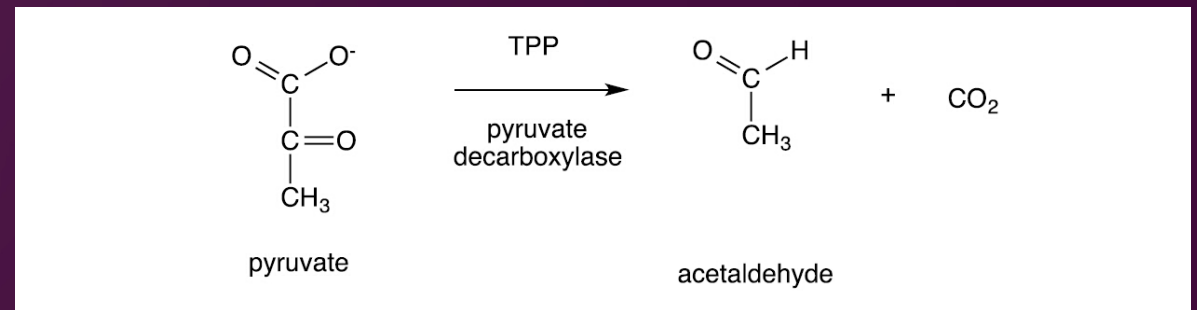


7 types (cont.)

Hydrolases



Lyases

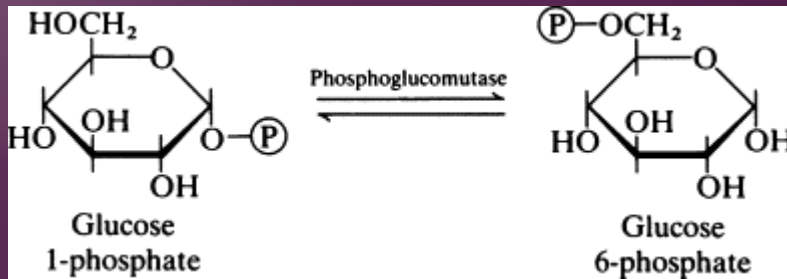


2.1: Enzymes as Biological Catalysts

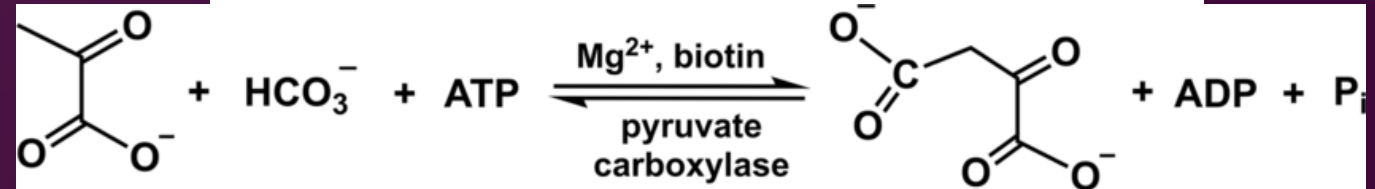
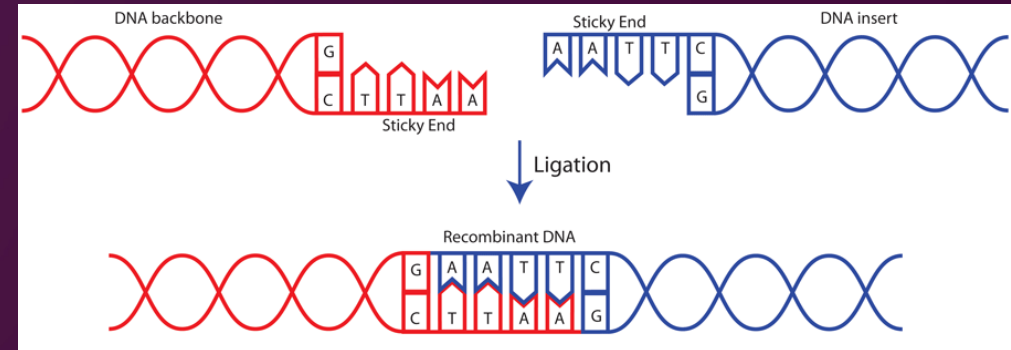


7 types (cont.)²

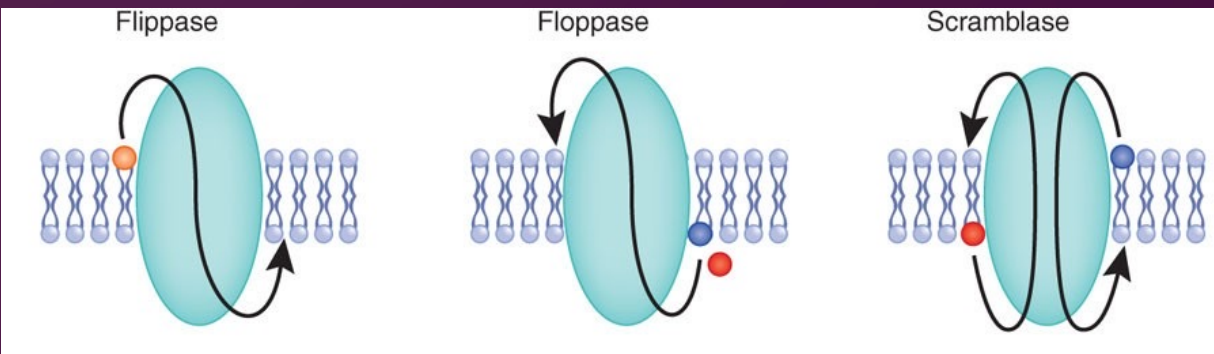
Isomerases



Ligases



Translocases



2.1: Enzymes as Biological Catalysts



Question

“...Niemann-Pick disease is caused by a deficiency in *sphingomyelin phosphodiesterase*, a lysosomal enzyme that cleaves the phospholipid sphingomyelin into ceramide and phosphocholine by introducing a molecule of water across the phosphate ester bond...”

What class of enzyme is sphingomyelin phosphodiesterase?

- A. Transferase
- B. Kinase
- C. Hydrolase
- D. Lyase



Question

In the biosynthesis of vitamin B6 (pyridoxine), a novel bacterial enzyme known as pyridoxine 5'-phosphate synthase (PdxJ) is responsible for a crucial step in the ring closure reaction. It catalyzes the complex conversion of 1-deoxy-D-xylulose 5-phosphate (DXP) and aminoacetone to form pyridoxine 5'-phosphate (PLP). A related enzyme in the pathway, pyridoxal reductase, utilizes a reduced nicotinamide cofactor to convert pyridoxal to pyridoxine.

Based on the passage information, what are the most likely products when pyridoxal reductase acts upon its substrate, pyridoxal, in the presence of its required cofactor?

- A) Pyridoxine, NAD^+
- B) Pyridoxine and NADH
- C) Pyridoxal 5'-phosphate and ATP
- D) Pyridoxamine and H_2O

